作业题 9A 7.3 7.4 7.6 7.9

9B 7.12 7.19

7.3 $\begin{cases} P_1 S h_1 = \nu_1 RT \\ P_2 S(h_1+\Delta h) = (\nu_{He}+\nu_1)RT \end{cases}$

$T=300K$

$S=2\times10^{-4}m^2$

$h_1=0.12m$

$P_0=1.013\times10^5 Pa$

He

$\Delta h=0.16m$

$\Rightarrow (P_2-P_1)Sh_1 + P_2 S\Delta h = \nu_{He}RT$

$P_2-P_1 = 160\times\frac{P_0}{760} = 2.13\times10^4 Pa$

$(P_2-P_1)S(h_1+\Delta h) = \nu_{He}RT$

$\Rightarrow \nu_{He} \approx 4.89\times10^{-4} mol$?

$m = \nu_{He}M = 1.96\times10^{-6} kg$

$N = \nu_{He}N_A = 3.03\times10^{20}$

$n = \frac{N}{S(h_1+\Delta h)} = 5.41\times10^{24}$

7.4 $2dN\mu v\sin\theta = pS\,dt$

$S=2\times10^{-4}m^2$

$v=500m/s$

$\theta=30°$

$n=10^{23}$

$\Rightarrow p = \frac{2\mu v\sin\theta}{S}\frac{dN}{dt} = 1.8\times10^4 Pa$

A

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$R=8.13\,J/(mol\cdot K)$

7.6 (1) $\varepsilon=\frac{3}{2}kT$

平动 $E=\varepsilon N_A=\frac{3}{2}RT=4878\,J$

转动 $E=RT=3252\,J$

内能 $E=\frac{5}{2}RT=8130\,J$

(2) 平动 $E=4878\,J$

转动 $E=0$

内能 $E=4878\,J$

7.9 $NE=\frac{3}{2}\nu R\Delta T$

$\Delta T=\frac{2NE}{3\nu R}=0.013K$

7.12 $pV=\nu RT$

$\Rightarrow \frac{RT}{M}=\frac{pV}{\nu M}=\frac{pV}{m}$ $T=\frac{pV}{\nu R}=307K$ | 0.29 kg | 0.1 mol |

$P=8.31\times10^4 P_0$

$V=3L$

$v_p=\sqrt{\frac{2RT}{M}}=\sqrt{\frac{2pV}{m}}=1579\,m/s$

$\bar{v}=\sqrt{\frac{8RT}{\pi M}}=\sqrt{\frac{8pV}{\pi m}}=1782\,m/s$ $\sqrt{\overline{v^2}}=\sqrt{\frac{3RT}{M}}=\sqrt{\frac{3pV}{m}}=1934\,m/s$

$f(v)=4\pi\left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}}e^{-\frac{Mv^2}{2RT}}v^2$ $v_0=1000\,m/s$, $\Delta v=1\,m/s$

$\frac{\Delta N}{N}=\int_{1000}^{1001}f(v)dv\approx f(v_0)\Delta v=2.38\times10^{-19}$

$\Delta N=2.38\times10^{19}$

7.19 $T=273K$, $M=0.0289\,kg/mol$

$\begin{cases}p=p_0e^{-\frac{mgh}{RT}}\\ p=\frac{3}{4}p_0\end{cases}\Rightarrow h=\frac{-RT\ln\frac{3}{4}}{Mg}=2209m$

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